



Submitted for:	1M1B AI for Sustainability Virtual Internship (In Collaboration with IBM SkillsBuild & AICTE)
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Code Repository:	github.com/SoumyadeepDas001/greenloop-ai
Live App:	greenloop-ai-flax.vercel.app

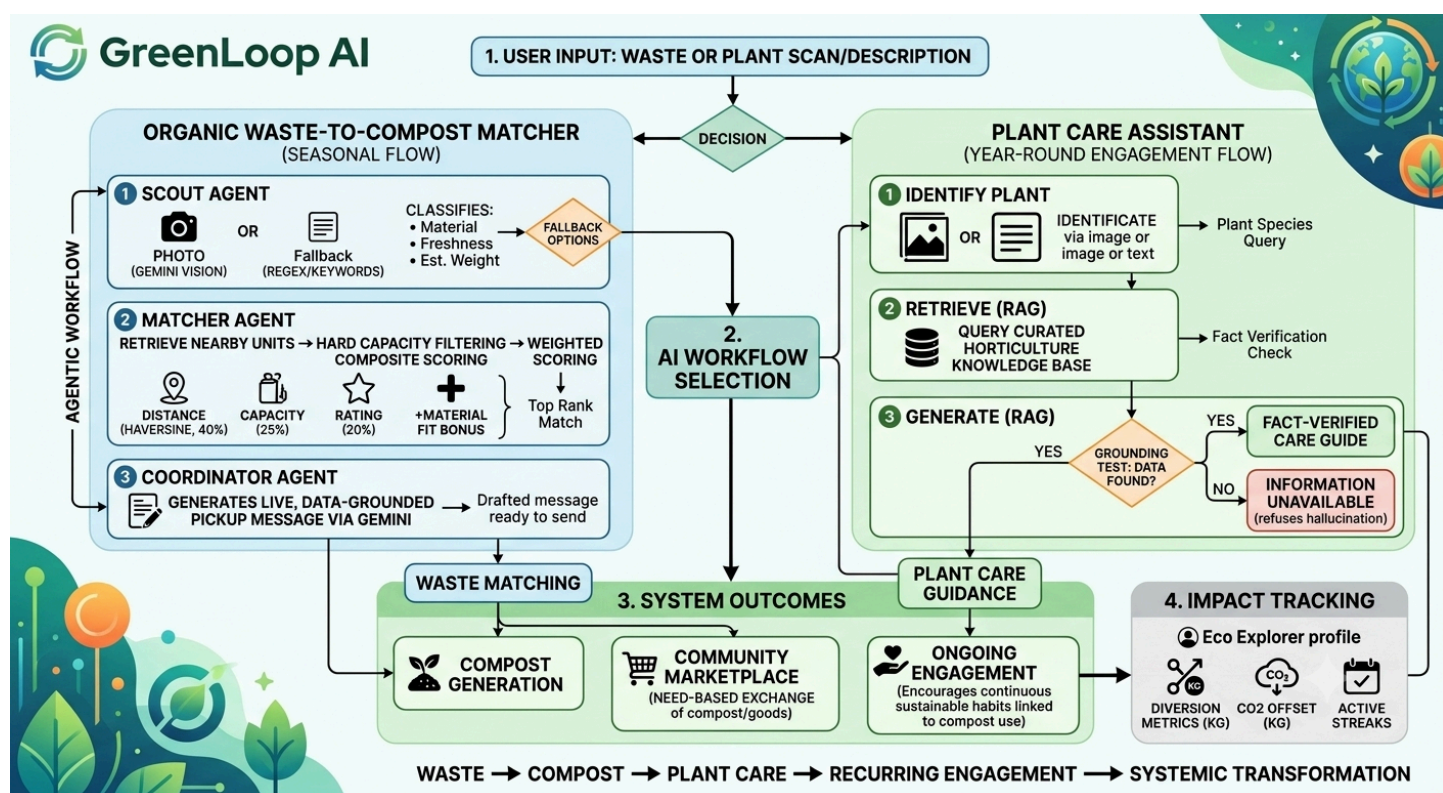
SDG Alignment

Primary	SDG 12 – Responsible Consumption and Production
Secondary	SDG 15 – Life on Land
Tertiary	SDG 11 – Sustainable Cities and Communities

Problem Statement

"How might we use AI to solve a problem that repeats every festival season across Indian cities: tons of flower and organic waste generated at temples, pandals, and community celebrations, with no structured way to route it to the composting units, incense makers, or urban farms that could actually use it? Right now, this waste either goes to landfill or depends on ad-hoc, manual coordination between event organizers and waste processors – a system that doesn't scale and breaks down the moment volume spikes during peak festival periods. How might we use AI to automatically classify this waste, match it in real time with the nearest capacity-aware composting resource, and turn the resulting compost into an ongoing reason for that same community to stay engaged with sustainable practices – so that landfill diversion stops being an exception and becomes the default outcome?"

AI Solution Overview



GreenLoop AI is built on a simple but deliberate premise: most sustainability tools solve a single waste problem and stop there. GreenLoop instead treats waste as the first step in a cycle, not an endpoint – closing a real loop instead of solving one problem in isolation: **Waste → Compost → Plant Care**, powered by two connected, fully-implemented AI flows that share a common architectural pattern rather than existing as separate, bolted-together features.

Waste-to-Compost Matcher – Agentic Workflow

Matching waste to the right destination isn't a single classification task – it's a sequence of decisions, each depending on the last. GreenLoop models this explicitly as a three-agent pipeline, where each agent has one clear responsibility and hands its output to the next:

- **Scout Agent** – classifies material type, freshness, and estimated weight, using Gemini Vision on photos or keyword/regex parsing on text descriptions when no image is available. This dual-path design means the system degrades gracefully instead of failing outright when a photo isn't provided.
- **Matcher Agent** – retrieves nearby composting units and ranks them using a weighted composite score: distance (Haversine great-circle formula, 40%), available capacity (25%), and rating (20%), with an additional material-fit bonus. Units without sufficient remaining capacity for the estimated waste weight are filtered out entirely before ranking – the system doesn't recommend a match it knows can't actually be fulfilled.
- **Coordinator Agent** – drafts a real pickup coordination message, generated live via the Gemini API and explicitly grounded in the actual classification and match data, rather than a static template. This means the message a user sends genuinely reflects what was detected and matched, not a placeholder.

This isn't three separate scripts – it's one reusable orchestration pattern, later extended to a second, structurally identical use case (matching students in need with available secondhand books and uniforms), proving the architecture generalizes rather than being hardcoded to one problem.

Plant Care Assistant – Retrieval-Augmented Generation (RAG)

Generic AI-generated care advice risks being subtly wrong – a hallucinated watering schedule can quietly kill a plant. GreenLoop's Plant Care Assistant is built specifically to avoid that failure mode:

- **Retrieve** – identifies the plant species, via image recognition or text input, and looks up its entry in a curated horticulture knowledge base rather than relying on the model's general training knowledge.
- **Generate** – produces a natural-language care guide using *only* the retrieved facts. This was directly verified, not just assumed: when asked for care information on a species with no data on file, the system correctly responded that the information was unavailable, rather than generating a plausible-sounding but fabricated answer. That single test result is the clearest evidence in this project that the Responsible AI commitment to transparency isn't just a line in a slide – it's enforced in the actual running system.

Additional AI components used

- **Multimodal inputs** – Gemini Vision handles real photo classification across both the compost and plant-care flows, with a text-based fallback ensuring the system remains usable even without a camera or clear image.
- **Entity extraction** – weight, freshness, and material type are parsed directly from free-text descriptions, turning unstructured user input into structured data the rest of the pipeline can act on.
- **Conversational interface** – a router-based chat assistant (Router Agent → PlantCare / MatchStatus / Fallback agents) handles open-ended questions, applying the same single-responsibility agent pattern used in the core matching pipeline rather than a separate, disconnected chatbot logic.
- **Prompt engineering** – every Gemini call uses a structured, explicitly grounded prompt designed to constrain the model's output to real, retrieved data rather than open-ended generation – the mechanism underlying every Responsible AI claim made elsewhere in this document.

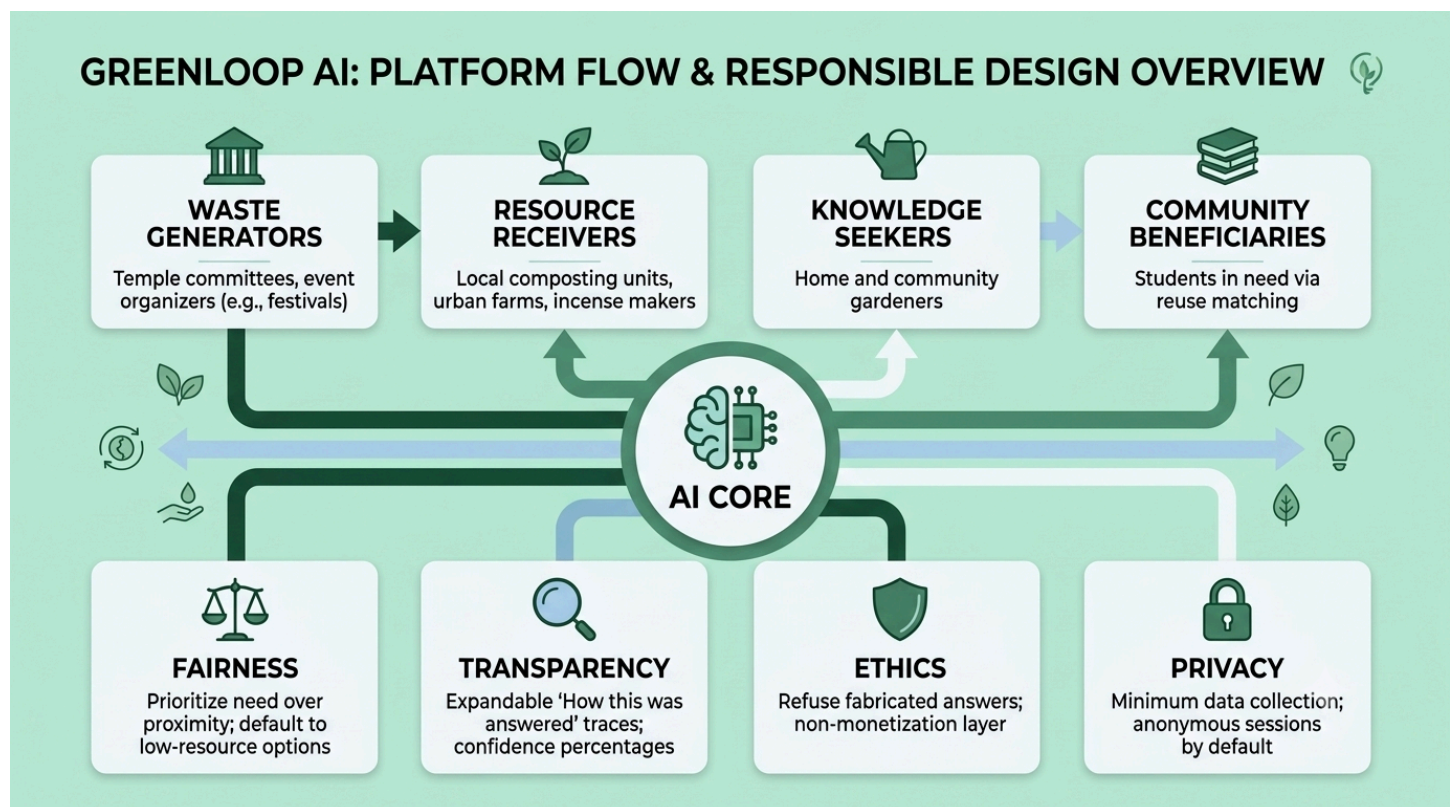
Target Users

Temple committees and event organizers – generate large volumes of flower and organic waste during festivals, currently with no structured way to route it anywhere but landfill; GreenLoop gives them a one-step way to divert it instead of absorbing that as an unavoidable cost of the event.

Local composting units, incense makers, and urban farms – currently rely on manual outreach or word-of-mouth to find waste sources; the platform gives them predictable, capacity-aware matched supply instead of unpredictable, ad-hoc pickups.

Home and community gardeners – want reliable plant care guidance but are underserved by generic advice that doesn't distinguish a real fact from a plausible-sounding guess; they get care instructions grounded in an actual knowledge base, with the system honest about what it doesn't know.

Students in need of secondhand books or uniforms – benefit from the same underlying matching engine applied to a second vertical, proving the architecture isn't single-purpose – need-based access to materials without added cost, prioritised by who needs it most rather than just who's nearest.



Responsible AI Considerations

Fairness

- Reuse matching prioritizes need over pure proximity, so the closest match isn't automatically the one selected if a farther user has a higher documented need.
- Care and craft suggestions default to low-resource options first, rather than assuming access to specialty tools, water-heavy setups, or paid materials – kept usable for whoever encounters the app, not just well-resourced users.

Transparency

- Every AI response includes an expandable "How this was answered" trace, showing which agent or retrieval step produced the answer – not a black box the user has to trust blindly.
- Every classification displays its confidence percentage rather than false certainty, and low-confidence results explicitly fall back to a "not sure – here's what to do instead" message rather than guessing with unwarranted authority.

Ethics

- No monetization layer exists on the exchange side – all matching is need-based, not commercial, to preserve free access rather than letting the platform quietly become a marketplace with pricing pressure.
- The system is designed to refuse fabricated answers over convenient ones: verified directly, asking about a plant with no data on file returns "not available" instead of an invented care guide – a deliberate ethical choice, not an accidental limitation.

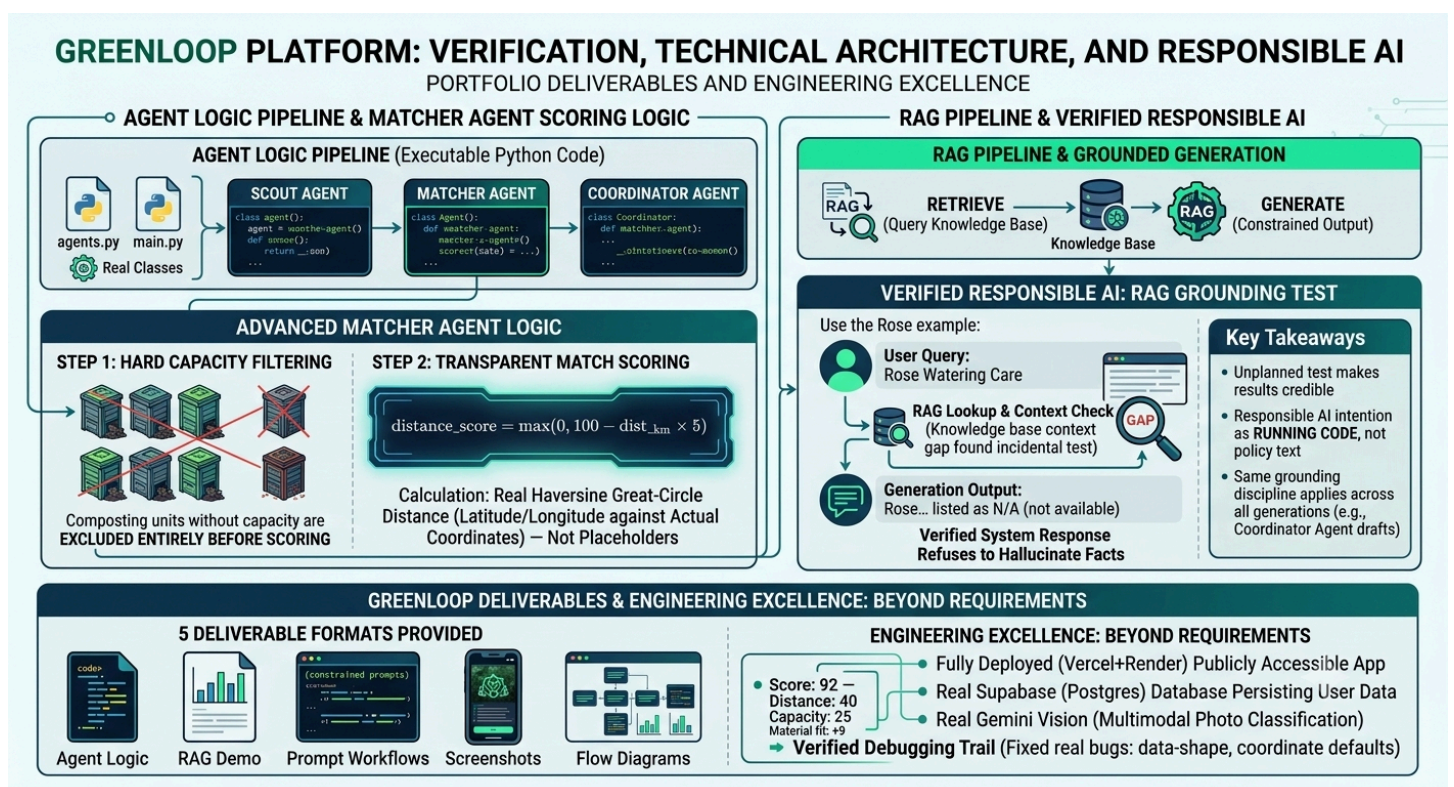
Privacy

- Only the minimum data needed for pickup coordination is collected – no personal identity is exposed between matched parties beyond what's necessary to complete a handoff.
- User sessions are anonymous by default, with no requirement to create an account or share personal information just to use the core matching features.

Expected Impact

Timeframe	What Changes	Who Benefits
Short-term	Festival organic waste gets diverted from landfill instead of left to rot uncollected; manual coordination between waste generators and composting units drops to near-zero, since matching happens automatically	Temple committees/event organizers (less waste to dispose of), composting units (predictable, matched supply instead of chasing leads)
Medium-term	The Plant Care module gives users a reason to return outside festival season – turning a one-time diversion action into a recurring habit rather than a single seasonal event	Home and community gardeners (ongoing, grounded care guidance tied to the compost they helped create)
Long-term	The same matching engine is designed to extend to secondhand-item reuse and inorganic waste routing (craft upcycling or correct recycling), without rebuilding the core architecture – proving the system scales in scope, not just usage	Students needing books/uniforms (Reuse Matcher), craft/recycling ecosystems, and the platform itself (one engine, multiple verticals, lower long-term maintenance cost)

Prototype & Demo



The guideline asks for any one of four demo formats. This project delivers all four, plus a fifth that goes beyond what was required:

- **Agent logic** – a fully working Python implementation (`agents.py`), runnable end-to-end via `python3 main.py`. Not pseudocode or a diagram description – real, executable agent classes (Scout, Matcher, Coordinator) that a reviewer can run themselves and watch produce different outputs for different inputs.
- **RAG demo** – the Plant Care Assistant, with a verified grounding test: asking about a species with no data on file returns "not available" instead of a fabricated answer, proving the retrieval-then-generate pipeline actually constrains the model rather than just claiming to.
- **Prompt workflows** – structured, explicitly grounded prompts used for every Gemini call, each designed to constrain the model's output to real, retrieved, or classified data rather than open-ended generation.
- **Screenshots** – shown below, captured directly from the live deployed application, not a mockup or design tool – what's pictured is what actually runs.
- **Flow diagram** – the agent pipeline (Scout → Matcher → Coordinator) and the RAG pipeline (Retrieve → Generate) are both documented as explicit sequential flows in the architecture section, satisfying the flowchart format as well.

Beyond the requirement:

A fully deployed, publicly accessible live application – Vercel (frontend) + Render (backend) – reachable by anyone with the link, not confined to a local machine or a single demo session.

A real Supabase (Postgres) backend, meaning user data, matches, and activity genuinely persist across sessions rather than resetting on every page load.

Real Gemini Vision integration, so waste and plant classification can run on an actual photo, not only a typed text description.

A verified debugging trail: this project was tested repeatedly, with real bugs found and fixed (data-shape mismatches, coordinate defaults, dependency conflicts) rather than presented as a single clean pass with no visible iteration.

Agent Scoring Logic (Matcher Agent)

```
total_score = (distance_score × 0.40)
              + (capacity_score × 0.25)
              + (rating_score   × 0.20)
              + material_fit_bonus (up to +15)
```

Where `distance_score = max(0, 100 - dist_km × 5)`, calculated using the real Haversine great-circle distance formula against actual latitude/longitude coordinates – not a placeholder or randomized value.

Two additional guarantees built into this logic, not just the formula itself:

- **Hard capacity filtering happens before scoring, not after** – any composting unit without enough remaining capacity for the estimated waste weight is excluded entirely, so the system never recommends a match it already knows can't be fulfilled.

- **Every score is broken down and shown to the user** – distance, capacity, and rating contributions are displayed separately (e.g., "Score: 92 – Distance: 40, Capacity: 25, Rating: 18, Material fit: +9"), not collapsed into a single opaque number, so the ranking logic is auditable rather than a black box.

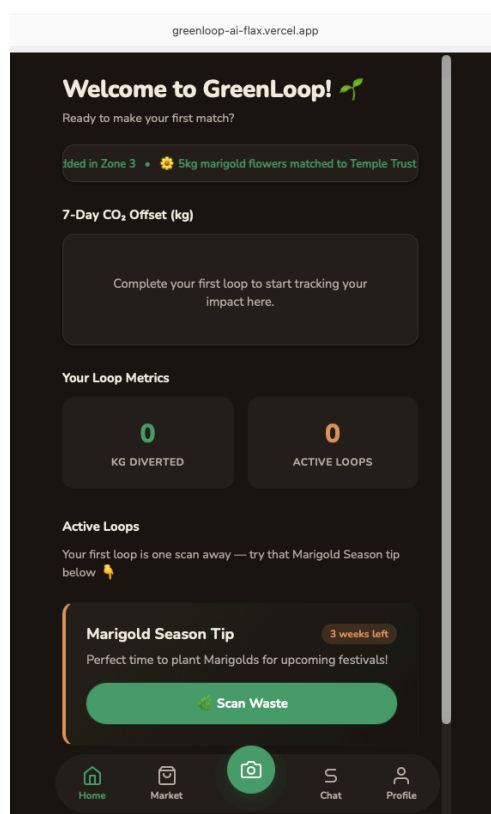
Verified Responsible-AI Behavior (RAG Grounding Test):

When asked for care information on a species with no watering data in the knowledge base, the system responded: *"Based on the provided context, the watering needs for a Rose are listed as N/A (not available)."* This confirms the RAG pipeline refuses to hallucinate facts outside its grounded knowledge base, rather than inventing a plausible-sounding but false answer.

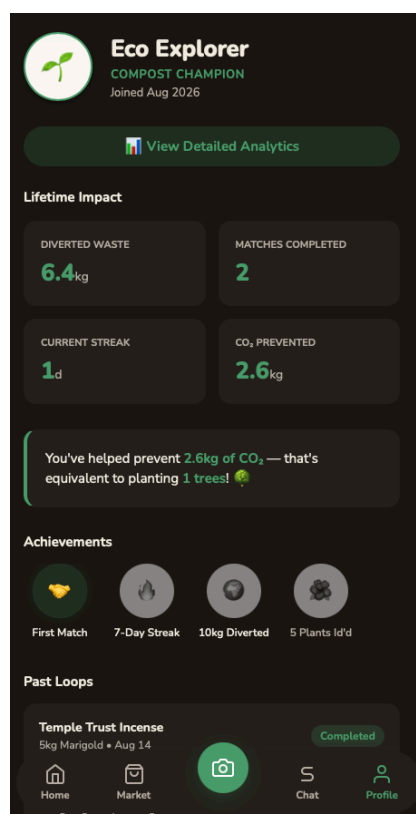
This matters beyond the single test case, for three reasons:

- **It was an unplanned test, not a scripted demo** – the gap in the knowledge base was discovered incidentally, not engineered in advance to showcase this behavior, which makes the result more credible, not less.
- **It demonstrates the Transparency and Ethics principles as running code, not policy text** – most Responsible AI sections describe an intention; this one shows the intention actually enforced at runtime.
- **The same grounding discipline applies across every generation step in the app** – the Coordinator Agent's drafted messages and the chat assistant's sub-answers are all built on the same "only state what was retrieved" pattern, not just the Plant Care flow.

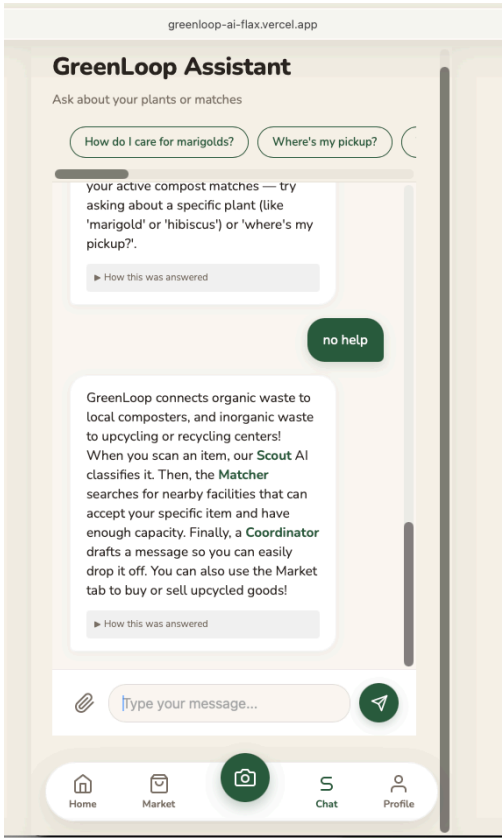
Live Application Screenshots



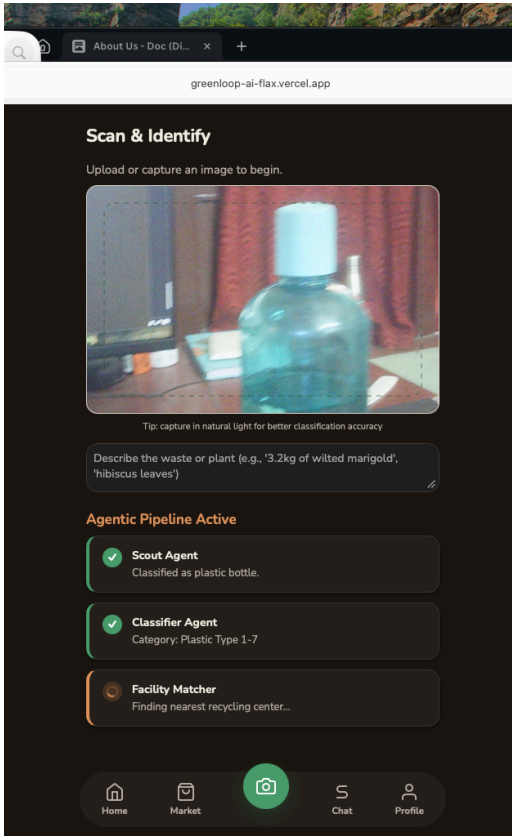
Home - GreenLoop



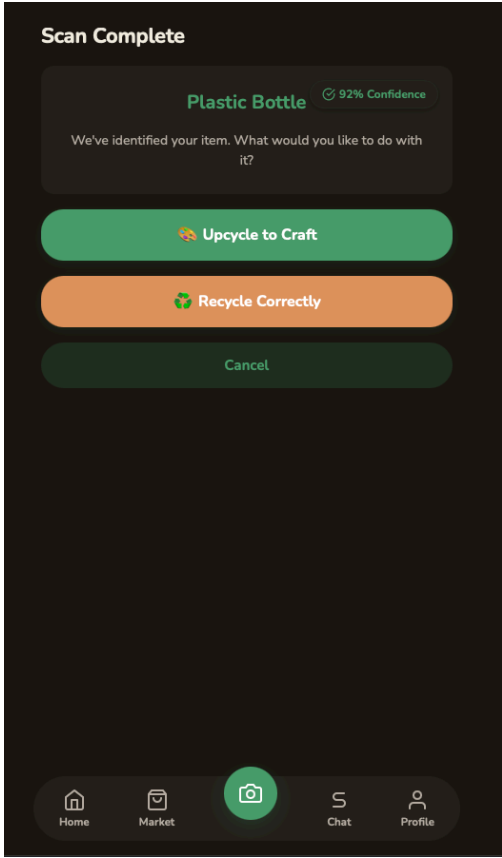
Profile - GreenLoop(After Loop)



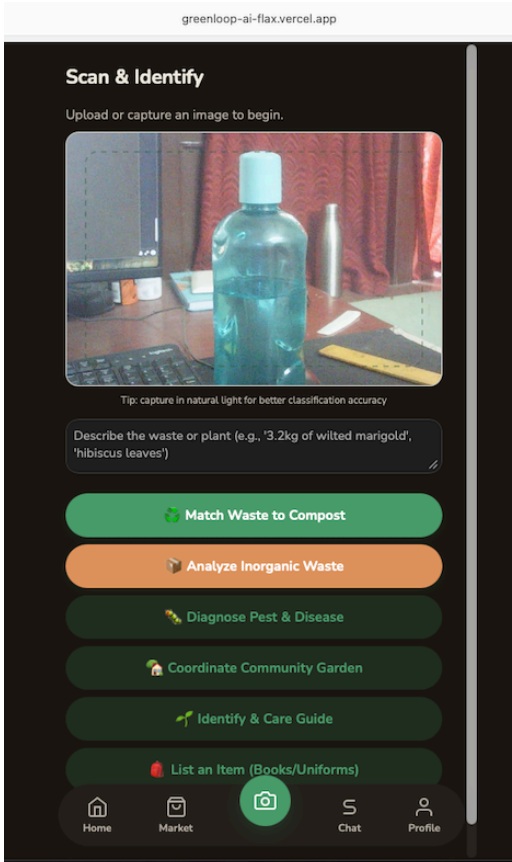
Greenloop Assistant



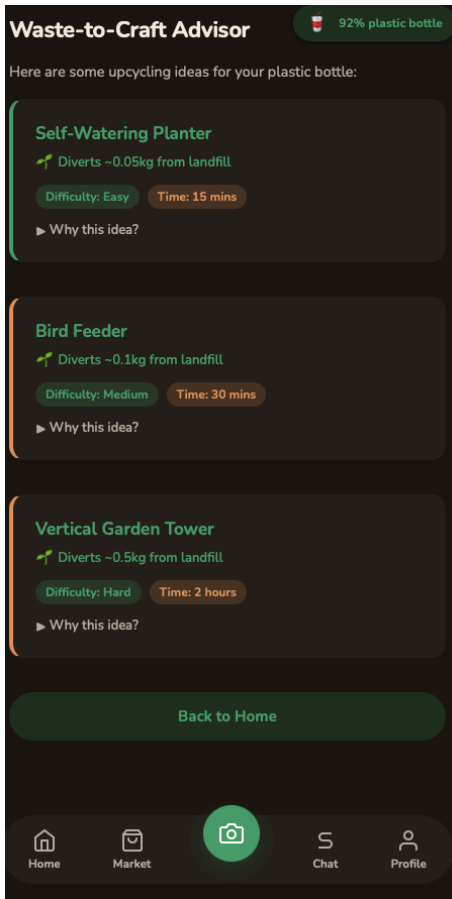
Live photo classification via Gemini Vision



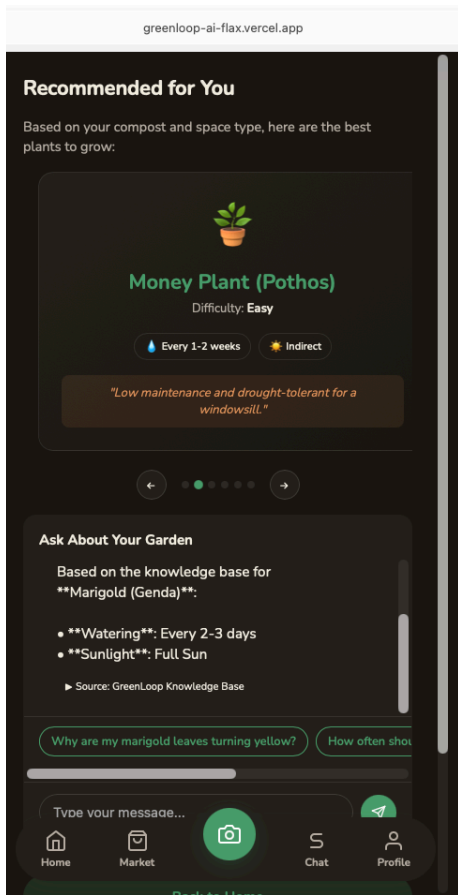
After Scan



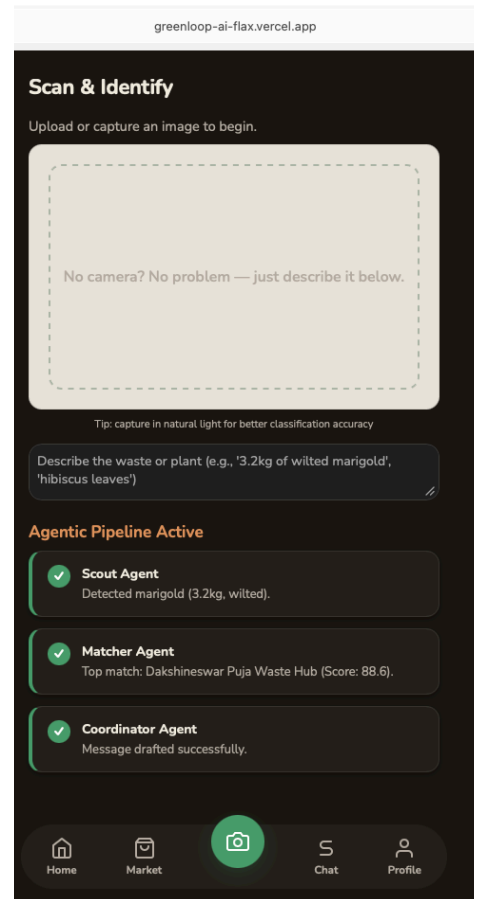
Before Scan



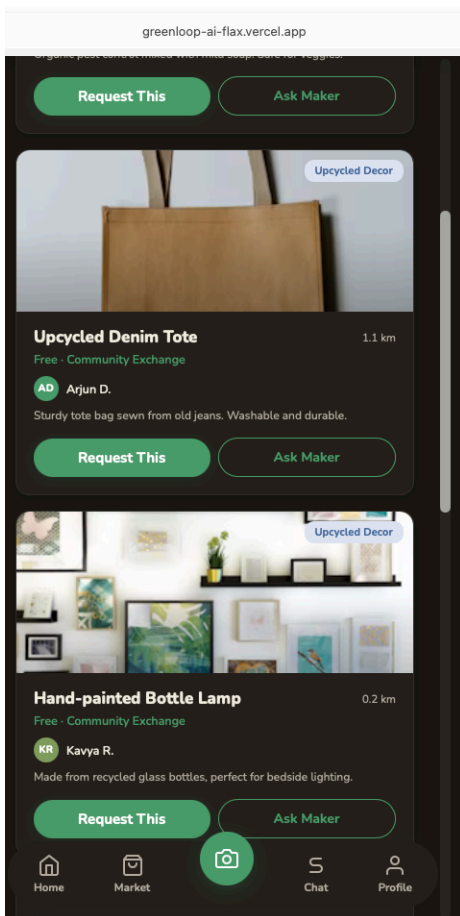
Craft Advisor



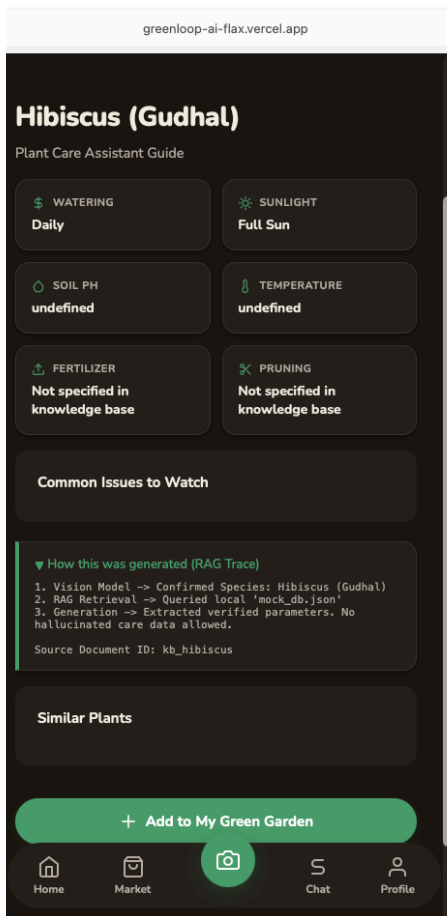
Recommendation



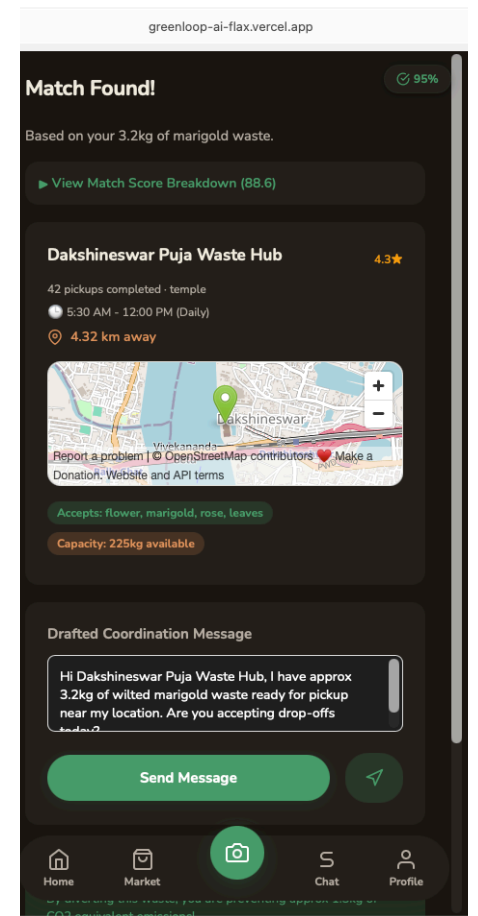
Description to Scan



Market Zone



Garden Maker



After Execution

Impact Statement

Systemic Transformation: What Changes if Implemented?

The implementation of GreenLoop AI transitions community waste management from a fragmented, manual process into a seamlessly automated, closed sustainability loop. Currently, festival and household organic waste lacks an accessible path to composting, leading to widespread landfill disposal. This solution removes that coordination barrier by automatically matching waste to capacity-aware local units.

Beyond immediate waste diversion, the platform establishes a long-term engagement model. By integrating a Plant Care Assistant powered by Retrieval-Augmented Generation (RAG), the system keeps users engaged year-round with trustworthy horticulture guidance connected directly to the compost they helped create. This dual approach effectively targets SDG 12 (Responsible Consumption and Production), SDG 15 (Life on Land), and SDG 11 (Sustainable Cities and Communities) by turning a one-time diversion action during festival season into a recurring, community-wide habit. Furthermore, the underlying agentic architecture is designed to scale beyond organic waste, creating a foundational engine for routing inorganic recycling and secondhand goods without requiring core system rebuilds.

Stakeholder Value: Who Benefits and How?

- **Waste Generators (Temples, Event Organizers, and Households):**

These groups gain a frictionless, automated method to divert organic waste—especially festival flower waste—away from landfills.

The burden of manual coordination is eliminated, as the AI Scout Agent automatically classifies the material's freshness and weight.

User privacy is maintained throughout the process, as only the minimum data required for pickup coordination is collected and personal identities are not exposed.

- **Composting Units, Incense Makers, and Urban Farms:**

Local processing facilities receive a predictable, highly suitable supply of organic materials without relying on manual outreach.

Supply is dynamically routed to them via the Matcher Agent, which optimizes logistics using a weighted score that accounts for their available capacity (25%), rating (20%), and actual distance calculated via the Haversine formula (40%).

- **Home and Community Gardeners:**

Plant growers receive highly accurate, grounded plant care guidance through the RAG-powered Plant Care Assistant.

Because the AI is restricted to a curated horticulture knowledge base, gardeners are protected from generic or hallucinated advice; if data is missing, the system explicitly states the information is unavailable.

Craft and care suggestions default to low-resource options, ensuring the guidance is equitable and accessible.

- **Students and Low-Resource Individuals:**

Vulnerable community members benefit directly from the platform's Reuse Matcher extension.

This system utilizes the same matching engine to provide access to essential secondhand items, such as books and uniforms.

The exchange prioritizes actual need over pure geographic proximity and operates entirely without a monetization layer, preserving free access for those who need it most.

